MIL-C-60315A(MU)
2 February 1973
SUPERSEDING
MIL-C-60315(MU)
2 February 1965

MILITARY SPECIFICATION

CHARGE, PROPELLING, M67, FOR 105MM HOWITZER LOADING, ASSEMBLING AND PACKING

1. SCOPE

1.1 This specification covers the loading, assembling and packing for one type of propelling charge designated as Charge, Propelling, M67, for Howitzer, M2A2, M103 and M137.

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids or request for proposals, form a part of this specification to the extent specified herein.

SPECIFICATION

FEDERAL

PPP-C-843	- Cushioning Material, Cellulosic

MILITARY

MIL-A-2550	- Ammunition and Special Weapons, General Specification for
MIL-C-45662	- Calibration System Requirements

STANDARDS

MILITARY

MIL-STD-105	- Sampling Procedures and Tables for Inspection by Attributes (ABC-STD-105)
MIL-STD-109	- Quality Assurance Terms and Definitions
MIL-STD-1235	- Single and Multilevel Continuous Sampling Procedures and Tables for Inspection by Attributes
MS 63053	- Drum, Metal Shipping and Storage 15 and 19 Gallon

FSC: 1315

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T00:28Z
Check the source to verify that this is the current version before use.

DRAWINGS

ARMY

9217087 -Marking Diagram and Sealing of Steel Drums for Shipment of Propellants, Bagged Charges
9205472 -Charge, Propelling, M67 Assembly

(Copies of specifications, standards, drawings and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer).

3. REQUIREMENTS

3.1 Material.-Materials shall be in accordance with applicable drawings and specifications.

3.2 Propelling Charge Assembly.-The propelling charge assembly shall comply with all requirements specified on Drawing (dwg.) 9205472 and with all requirements specified in applicable specifications.

3.3 Increment Loading Assembly Weight.-The net weight of the propellant shall not differ by more than the following amount from that specified in the loading authorization:

Increment Number	Tolerance Ounces
1	0.02
2	0.015
3	0.015
4	0.015
5	0.015
6	0.02
7	0.02

3.4 Charge, Propelling Assembly Weight.-The charge, propelling assembly weight shall not differ by more than 0.10 ounces from the total of the amount specified in the loading authorization, plus tare weight of the bags and cord.(see 6.2).

3.5 Workmanship.-All parts and assemblies shall be fabricated and finished in a thorough, workmanlike manner. All parts shall be free of dirt, grease, and other foreign material. All required marking shall be neat and sharply defined.

4. QUALITY ASSURANCE PROVISIONS

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T00:28Z
Check the source to verify that this is the current version before use.

4.1 Responsibility for inspection.-Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements. Reference shall be made to Standard MIL-STD-109 in order to define the terms used herein. The provisions of Specification MIL-A-2550 shall apply.

4.1.1 Submission of product.-At the time each completed lot of items deliverable under the contract is submitted to the Government for acceptance, the contractor shall supply the following information accompanied by a certificate which attests that the information provided is correct and applicable to the product being submitted:

a. A statement that the lot complies with all of the quality assurance provisions specified in this specification.

b. Drawing and specification number and date, together with identification and date of changes thereto.

c. A statement that all material purchased by the contractor meets requirements, when such material is controlled by Government or commercial specifications referenced in any of the contractual documents, and the certificates of conformance are on file and available for review.

d. Number of items in the lot.

e. Date submitted.

The certificate shall be signed by a responsible agent of the certifying organization. The initial certificate submitted shall be substantiated by evidence of the agent's authority to bind his principal. Substantiation of the agent's authority will not be required with subsequent certificates unless, during the course of the contract, this authority is vested in another agent of the certifying organization.

4.2 Inspection Provisions

4.2.1 Lot formation.-The term "inspection lot" as used in this specification is defined as an essentially homogeneous collection of units of product from which a representative sample is drawn and inspected to determine conformance with applicable requirements. The sample selected shall represent only that quantity of units from which the sample was drawn and shall not be construed to represent any prior or subsequent quantities presented for inspection. Homogeneity shall be considered to exist provided the inspection lot has been produced by one

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T00:28Z
Check the source to verify that this is the current version before use.

manufacturer, in one unchanged process, in accordance with the same drawing, same drawing revision, same specification and same specification revision and complies with the provisions for submission of product as specified in MIL-STD-105. Changes to the process, specification, or drawing not affecting safety, performance, interchangeability, or storage, as determined by the Government, shall not be deemed to alter the homogeneity of an inspection lot. All materials submitted for inspection in accordance with this specification shall comply with the homogeneity criteria specified herein regardless of the type of sampling procedure which is being applied to determine conformance with requirements. Lot numbering, as required, shall be in accordance with MIL-STD-1168. Each lot shall contain:

a. Single perforated propellant from not more than one lot.
b. Multi perforated propellant from not more than one lot.

4.2.2 Examination.-The examinations listed in the following subparagraphs shall be performed on inspection lots (4.2.1) of the components or assemblies designated in the applicable paragraph headings. Inspection for critical defects (and major defects, when so specified) shall be 100 percent. Sampling plans for major and minor defects shall be in accordance with MIL-STD-105 except that continuous sampling plans in accordance with MIL-STD-1235 may be used if approved by the Government. Instead of applying the AQL's designated herein to the groups of major and minor defects listed, the contractor may elect to apply AQL's to individual defects using an AQL of 0.40 percent for each major defect and an AQL of 0.65 percent for each minor defect. When this method is elected, it shall be applied to all of the major and minor defects listed in the applicable subparagraphs of the section, except where 100 percent inspection is specified, and be documented as part of the contractor's inspection system. Equipment necessary for the performance of the inspections listed shall be in accordance with 4.2.4.

4.2.2.1 Bag increment (see dwg. 9205470 covering a detail of dwg. 9205472.

Categories	Defects	Method Of Inspection
Critical:	None defined	
Major:	None defined	
Minor:	AQL 1.5 percent	
	201. Any seam or opening incompletely stitched...	Visual

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T00:28Z
Check the source to verify that this is the current version before use.

202. Marking misleading or unidentifiable.....Visual
203. Foil missing or improperly assembled (applicable to increment 5 only).....Visual
204. Evidence of poor workmanship (see 3.5).....Visual

4.2.2.2 Increment loading assembly (see dwg. 9205471, covering a detail of dwg. 9205472.

Categories	Defects	Method Of Inspection
Critical:	None defined	
Major:	AQL 0.40 percent	
101.	Assembly damaged to the extent that propellant can escape.....	Visual
Minor:	AQL 1.5 percent	
201.	Any seam or opening incompletely stitched...	Visual
202.	Thread broken.....	Visual
203.	Evidence of poor workmanship (see 3.5).....	Visual

4.2.2.3 Charge assembly (see dwg. 9205472)

Categories	Defects	Method Of Inspection
Critical:	None defined	
Major:	AQL 0.40 percent	
101.	Assembly damaged to the extent that propellant can escape.....	Visual
Minor:	AQL 2.5 percent	
201.	Increments insecurely tied at junctions.....	Visual
202.	Ties loose, broken or insecurely tied.....	Manual-Visual
203.	Increment missing or improperly assemblied..	Visual
204.	Marking misleading or unidentifiable.....	Visual
205.	Evidence of poor workmanship (see 3.5).....	Visual

4.2.2.4 Drum, prior to filling (see dwg. MS 63053)

Categories	Defects	Method Of Inspection
Critical:	None defined	
Major:	AQL 0.40 percent	
101.	Foreign material in drum.....	Visual
Minor:	None defined	

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T00:28Z
Check the source to verify that this is the current version before use.

4.2.2.5 Drum sealed (see dwg. MS 63053 and 9217087)

Categories	Defects	Method Of Inspection
Critical:	None defined	
Major:	AQL 0.40 percent	
101.	Gasket missing or improperly positioned	Visual
102.	Locking ring improperly positioned	Visual
Minor:	AQL 1.0 percent	
201.	Car seal missing or improperly positioned	Visual
202.	Marking misleading or unidentifiable	Visual

4.2.2.6 Fiber Container

Categories	Defects	Method Of Inspection
Critical:	None defined	
Major:	AQL 1.0 percent	
101.	Gasket missing or incomplete	Visual
102.	Locking device damaged or improperly closed	Visual
103.	Holes or breaks in cover or body	Visual
104.	Damage to coating or cover	Visual
105.	Foreign matter or corrosion in interior	Visual
Minor:	AQL 1.0 percent	
201.	Marking missing, misleading or unidentifiable	Visual
202.	Interior or exterior torn or delaminated	Visual

4.2.3 Testing

4.2.3.1 Increment Loading Assembly weight (see 3.3) Major defect-The increment shall be weighed and checkweighed 100 percent. Any increment which fails to comply with the applicable requirements shall be classed defective and removed from the lot. Inspection equipment shall be in accordance with 4.2.4.

4.2.3.1.1 Method 1-The propellant shall be weighed in a container and placed in the bag, the bag sewn and the increment loading assembly weighed. The net weight of the propellant may be calculated using an average tare weight of the empty bags or the tare weight as marked on the individual bag.

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T00:28Z
Check the source to verify that this is the current version before use.

4.2.3.1.2 Method 2-The propellant in a container shall be weighed and then check weighed on a different balance and if performed manually, by an other operator. The propellant shall then be placed in the bag and the bag sewn. The weight of the loaded bags should be checked as a major defect using a sampling plan from MIL-STD-105 or MIL-STD-1235 with an AQL of 0.40 percent. The net weight of the propellant shall be calculated using an average weight of the empty bag or the tare weight as marked on the individual bag.

4.2.3.2 Air pressure test (see 5.1.3) Major defect.-The containers shall be checked 100 percent as specified in 4.3.1. If any container fails to meet the requirement it shall be rejected and removed from the lot. Test equipment shall be in accordance with 4.2.4 (applicable to steel drums only).

4.2.3.3 Charge, Propelling, Assembly Weight. (see 3.4) The defects shall be classified as follows: Critical defect: Weight deviation in excess of 3 times the requirement of 3.4. Major defect: weight deviation in excess of that permitted by 3.4 but not more than 3 times that permitted by 3.4. The assembly shall be weighed 100 percent and compliance with 3.4 shall be determined by using the average weight of the empty bag assembly. If the charge is outside the established weight tolerance remove increment 5 by raising it off the scale without breaking the cord and replace it with a weight representing the average weight of increment 5. If the weight is incorrect reject the charge. If the weight is correct and if it can be determined that the off weight of increment 5 is caused by the lead foil (tare weight of the bag) the charge shall be considered acceptable.

4.2.4 The inspection equipment required to perform the examinations and tests prescribed in this section is identified under the "Method of Inspection" headings in 4.2.2 and 4.2.3 see 6.3.

4.2.4.1 Inspection Equipment Capability and Accuracy.-All inspection equipment and facilities shall be of adequate quality and quantity to perform properly and accurately all the required inspections, measurements and tests. The inspection equipment shall be maintained in satisfactory condition for the continued proper performance of these functions and shall be calibrated in accordance with the provisions of MIL-C-45662.

4.2.4.2 Government Rights to Documentation.-Inspection equipment drawings and lists provided and revised in accordance with the requirements may be used by DOD activities for design, procurement, manufacture, testing, evaluation, production and receiving inspection, overhaul, shipping, storage, identifica-

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T00:28Z
Check the source to verify that this is the current version before use.

tion of stock, maintenance of equipment, construction, survey and whenever inspection equipment drawings are needed.

4.3 Test methods and procedures

4.3.1 Air pressure test.-The container shall be subject to an internal pressure of 2 to 3 pounds per square inch (p.s.i.) by a method satisfactory to the contracting officer's representative. A gage (designed to read 1/4 p.s.i.) or a water manometer shall be assembled in the system. A drop of 1/4 p.s.i. on the gage or 0.7 inches on the manometer in 1 minute shall be cause for rejection.

5. PREPARATION FOR DELIVERY

5.1 Packing

5.1.1 Level A.-The propelling charges shall be packed in containers conforming to Standard MS 63053-2. The gross weight of the pack shall not exceed 150 pounds. Cushioning material conforming to PPP-C-843 shall be added as required to provide a tight pack.

5.1.2 Level C.-The charges shall be packed in a fiber drum certified in accordance with ICC Specification 21C and meeting the following requirements. Size shall be 15-1/2 ± 1/2 inches in diameter by 26 ± 1 inches deep. A layer of aluminun foil shall be laminated into the body and the bottom if it is made of fiberboard. The bottom shall be either steel or asphalt laminated fiberboard. The bottom and top edge of the drum shall have 24 gauge welded galvanized steel chimes. The cover shall be either plain or interlocking (with Bottom) design lever lock type with gasket and shall provide a mositure-proof seal. The cover shall be made with either a galvanized finish or a baked on varnish finish resistant to chipping, peeling and cracking. The drum shall be tested in accordance with ICC Specification 21C except the drum shall be filled with not less than 250 pounds of dry sand for the drop test. The exterior of the drum shall be coated with a water-resistant material.

5.1.3 Air pressure test of container.-Immediately prior to packing,the containers shall be tested with compressed air, at an internal air pressure of 2 to 3 p.s.i. The containers shall withstand the specified pressure for one minute minimum witout leakage.

5.2 Marking

5.2.1 Steel Drums.-Marking containers shall be marked in

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T00:28Z
Check the source to verify that this is the current version before use.

accordance with drawing 9217087. The ICC nomenclature shall be "PROPELLANT EXPLOSIVES CLASS A".

5.2.2 Fiber Drums.-Marking Containers shall be marked in accordance with drawing 9217087. The ICC nomenclature shall be "PROPELLANT EXPLOSIVES CLASS B".

6. NOTES

6.1 Ordering data.-Procurement documents shall specify title, number and date of this specification.

6.2 Charge, Propelling Assembly Weight.-This requirement applies to the load plant only due to change in tare weight of the bags because of variation in bag material and moisture content.

6.3 Inspection Equipment Designs.-Inspection equipment designs are of two types-Government Special Inspection Equipment (SIE) designs and contractor designs. SIE designs, if available, are designated by drawing numbers (on the Equipment Lists referenced on the Equipment Tabulation) or (on the Equipment Lists referenced on the Equipment Tabulation) or (under the "Method of Inspection" headings in Section 4). Design responsibility for all other inspection equipment is assigned to the contractor. Equipment to be designed by the contractor is designed "CD". The contractor need not furnish any design when a complete Government SIE design is provided. Unless otherwise specified, however, the contractor may submit alternate or modified contractor designs of SIE in accordance with 6.3.2 and 6.3.3 should he elect to do so.

6.3.1 SIE designs.-SIE designs may consist of any of the following:

a. Detailed drawings which completely depict all information necessary for the fabrication and use of the item of inspection equipment.

b. A source control drawing or a specification control drawing as defined in MIL-STD-100.

c. An envelope drawing, as defined in MIL-STD-100, which established the criteria which a detail design shall meet. When envelope drawings are specified, the contractor shall prepare designs which comply with the criteria therein.

6.3.2 Contractor designs.-Contractor designs are required for all inspection equipment for which SIE designs are not specified and may include commercial equipment which the contractor proposes to use. (Commercial equipment is defined as unmodified equipment which is cataloged and available for

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T00:28Z
Check the source to verify that this is the current version before use.

purchase by the general public). Contractor designs shall include appropriate operating instructions, calibration procedures and maintenance procedures. Commercial equipment shall be fully described by catalog listings or other means which provide sufficient information to permit identification and evaluation by the Government and may include illustrations and engineering data. Designs shall be prepared for any special fixture(s) required to be used with commercial, or with SIE designs if not otherwise covered thereby (see 6.3.1c). Designs shall be of the category and form (per MIL-D-1000) specified in the Contract Data Requirements Lists (DD Form 1423). The contractor is referred to MIL-HDBK-204, "Inspection Equipment Design" for guidance. The specification number, paragraph number and defect number if such number has been assigned shall be referenced on each contractor design together with the component or assembly drawing number and revision letter to which the specific design applies.

6.3.3 Submission of designs for approval.-Contractor designs shall be approved by the Government prior to fabricating or procuring the equipment. Designs shall be submitted for approval in accordance with the stipulations, time frame and distribution specified in the Contract Data Requirements List (DD Form 1423) or in the contract. Partial submission of inspection equipment designs is permissible and encouraged. However, the completion date for design review will be based on the date of the final submission of designs. Picatinny Arsenal designs review will normally be accomplished within one month after receipt.

Custodian:
ARMY-MU

Preparing Activity
ARMY-MU

Project Number: 1315-A012

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T00:28Z
Check the source to verify that this is the current version before use.